

Reranking

Finding the right page, then putting it first

Weekend 5 · Retrieval Augmented Generation

Carlos Cotrini





Section 1

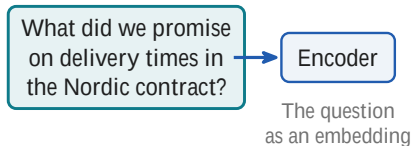
From retrieval to reranking

1

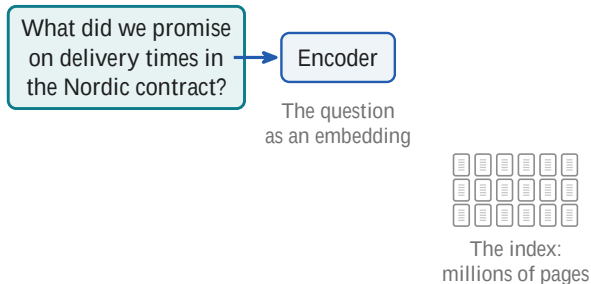
Where this morning le□ us

What did we promise
on delivery times in
the Nordic contract?

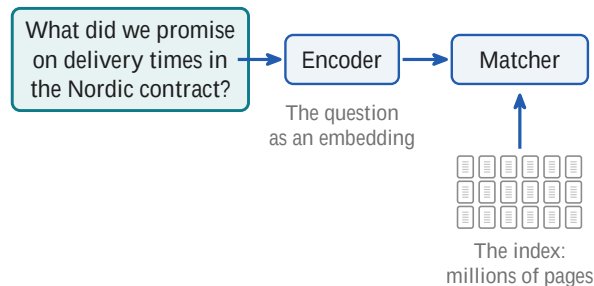
Where this morning le□ us



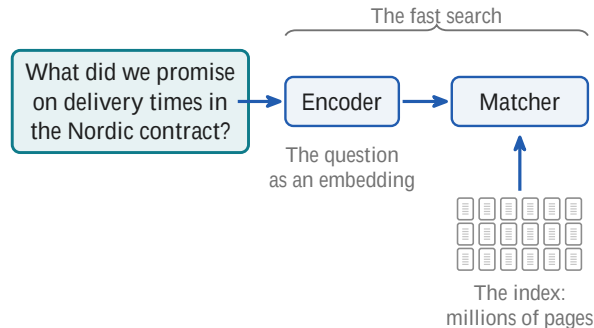
Where this morning le□ us



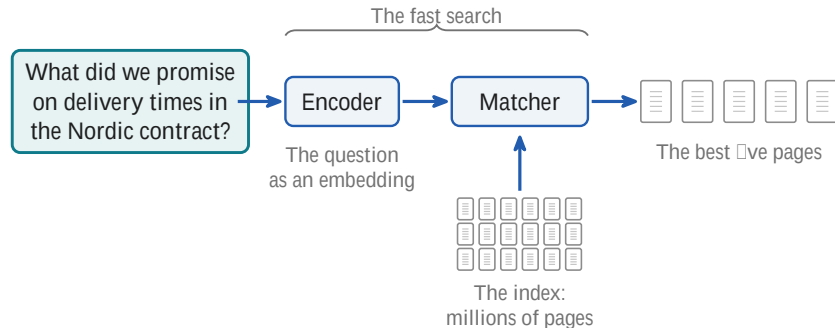
Where this morning le□ us



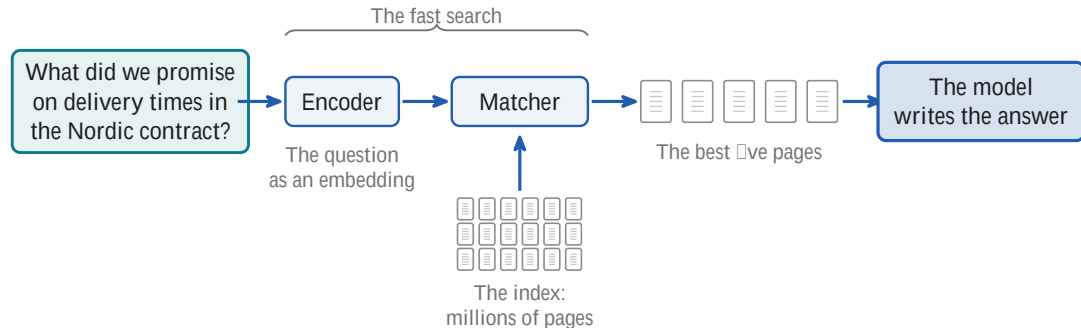
Where this morning le□ us



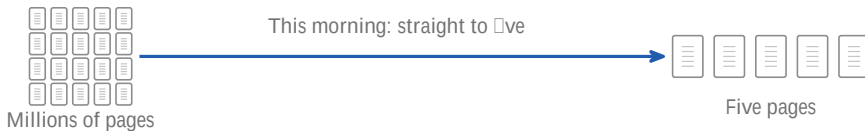
Where this morning le us



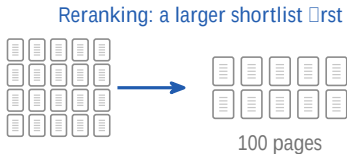
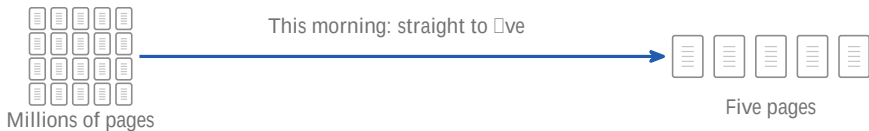
Where this morning le□ us



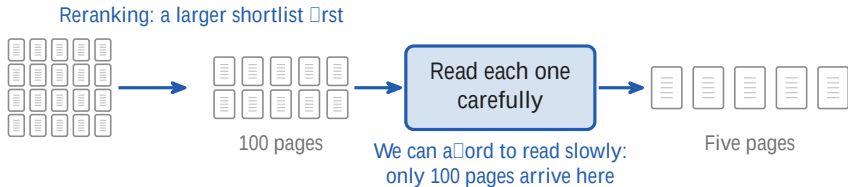
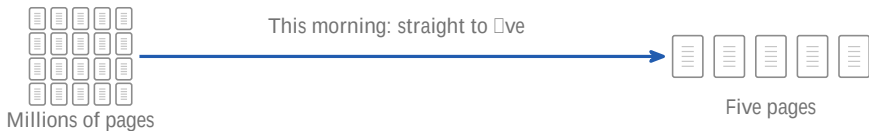
Reranking: choose from a larger shortlist



Reranking: choose from a larger shortlist



Reranking: choose from a larger shortlist

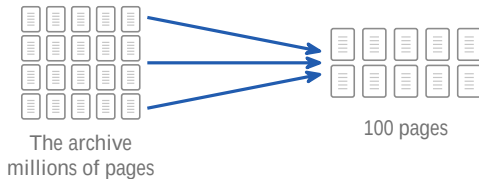


Two different jobs



The archive
millions of pages

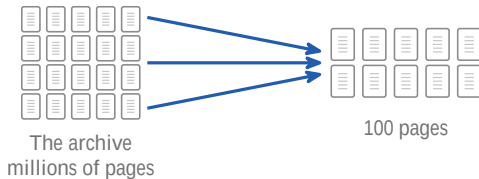
Two different jobs



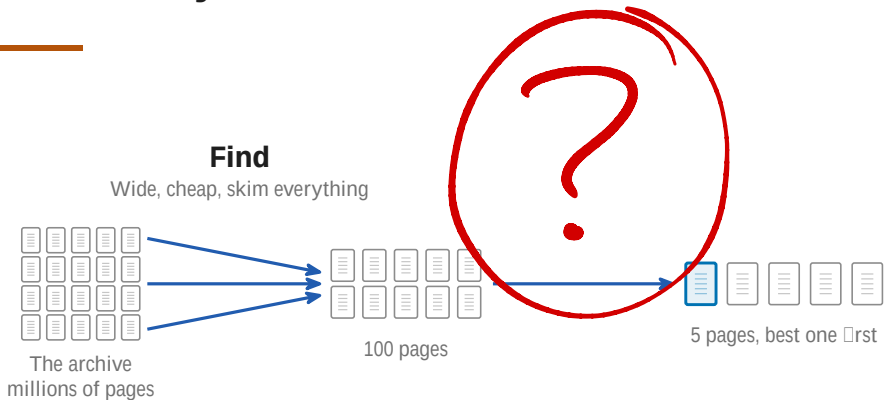
Two different jobs

Find

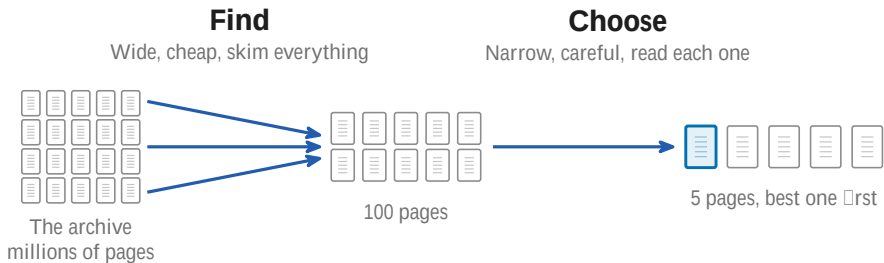
Wide, cheap, skim everything



Two different jobs



Two different jobs





Section 2

Three families of reranker

2

Three ways, cheapest first

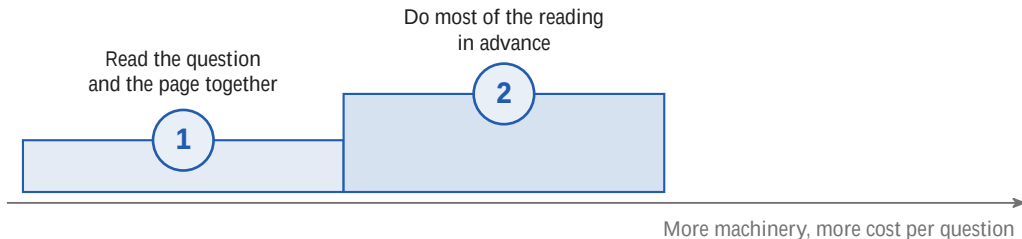
The 100 pages from the Nordic question, ordered three ways

Read the question
and the page together

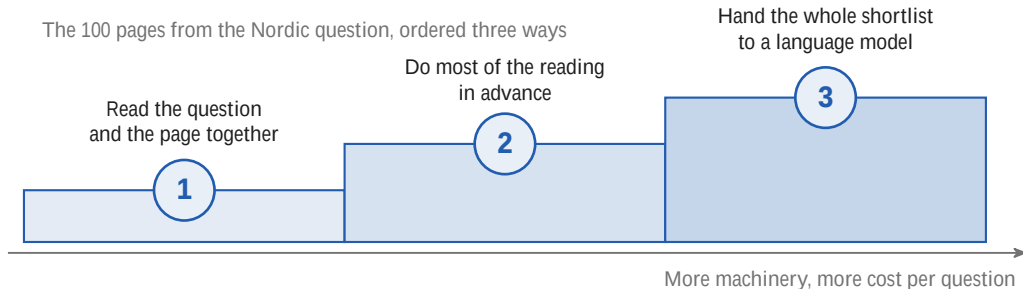


Three ways, cheapest first

The 100 pages from the Nordic question, ordered three ways



Three ways, cheapest first



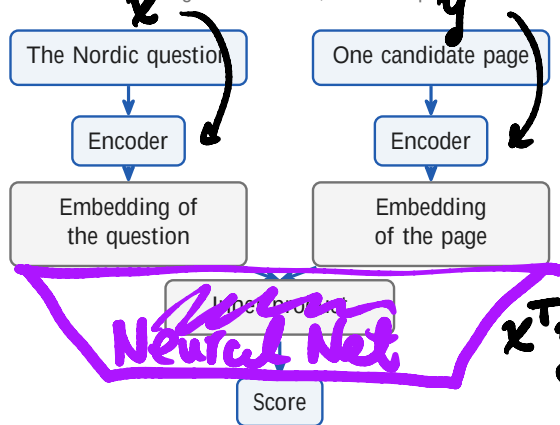
Rung 1 of three

1

**Read the question and
the page together**

Read the question and the page together

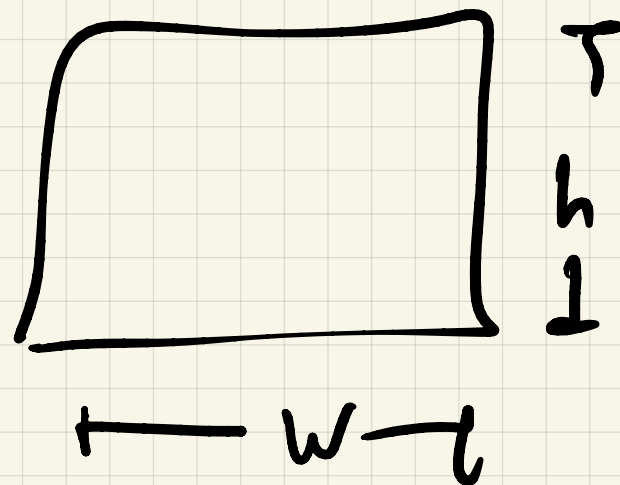
This morning: two encoders, one inner product



$$x = (x_1, x_2, \dots, x_{1536})$$

$$y = (y_1, y_2, \dots, y_{1536})$$

$$x^T y = x_1 \times y_1 + x_2 \times y_2 + \dots + x_{1536} \times y_{1536}$$

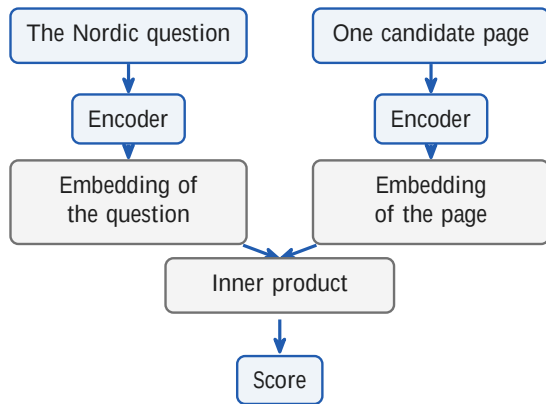


$$S = \underline{4h} + \underline{3w}$$

$$S = 5 \underline{h} \underline{w}$$

Read the question and the page together

This morning: two encoders, one inner product

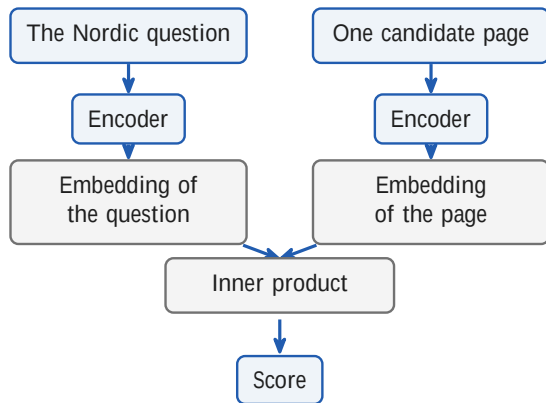


Why must the score be an inner product?

$$x \cdot y$$
$$x^T y$$
$$\langle x, y \rangle$$

Read the question and the page together

This morning: two encoders, one inner product

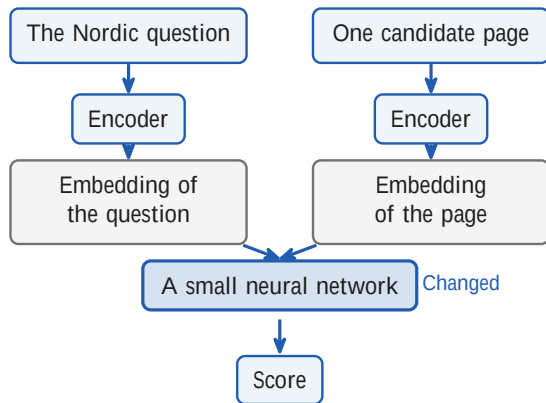


Why must the score be an inner product?

A neural network can approximate any function. Let it compute the score.

Read the question and the page together

This morning: two encoders, one inner product

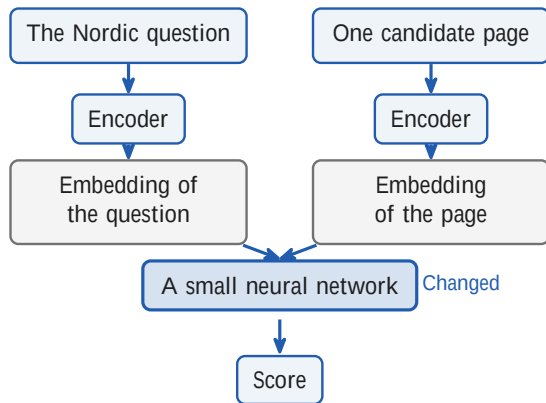


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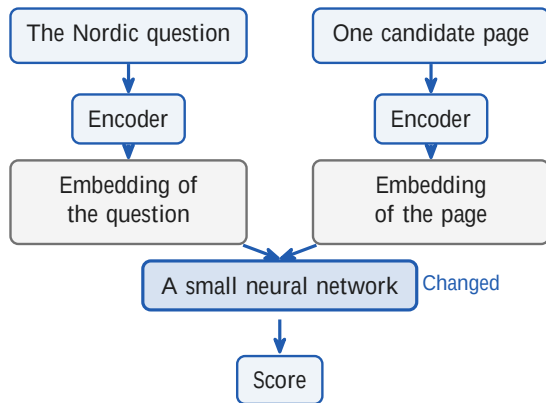
Why must the score be an inner product?

A neural network can approximate any function. Let it compute the score.

Why encode the question and the page apart?

Read the question and the page together

This morning: two encoders, one inner product



Why must the score be an inner product?

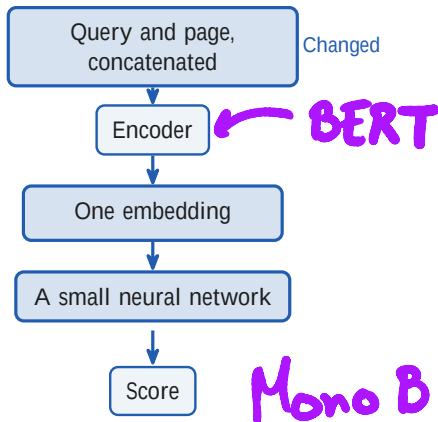
A neural network can approximate any function. Let it compute the score.

Why encode the question and the page apart?

Encoded together, one embedding can carry how the two relate, not only what each says alone.

Read the question and the page together

One encoder, reading both at once



Why must the score be an inner product?

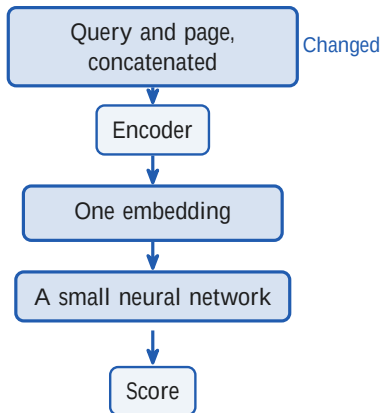
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Read the question and the page together

One encoder, reading both at once



Why must the score be an inner product?

A neural network can approximate any function. Let it compute the score.

Why encode the question and the page apart?

Encoded together, one embedding can carry how the two relate, not only what each says alone.

This is the cross encoder. One model run for every candidate page.

Rung 2 of three

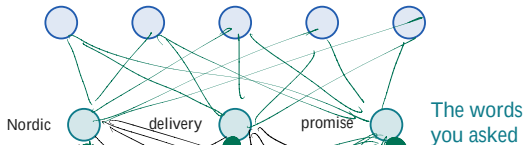
2

**Do most of the reading
in advance**

Best match, one question word at a time

A page of the parking policy

parking permit garage badge renew



A page of the Nordic contract

Collect all query words
into Q

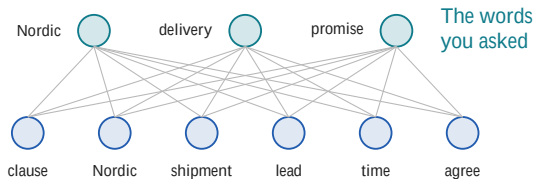
Collect all value words
into V

for each $q \in Q$
find the most sim
word $v_q \in V$

$$\text{ColBERT score} = \sum_{q \in Q} q^T v_q$$

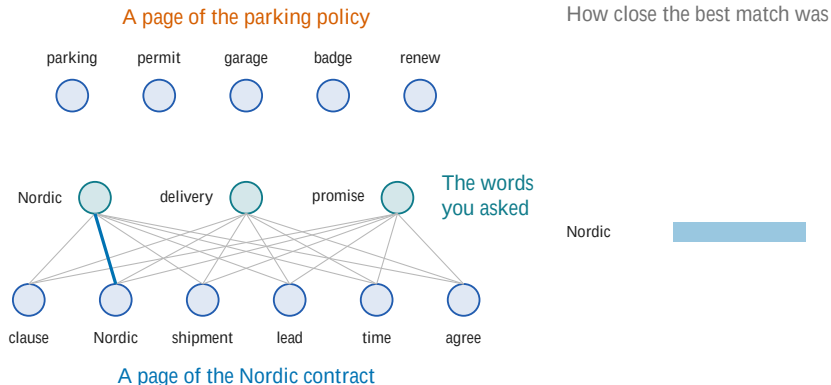
Best match, one question word at a time

A page of the parking policy

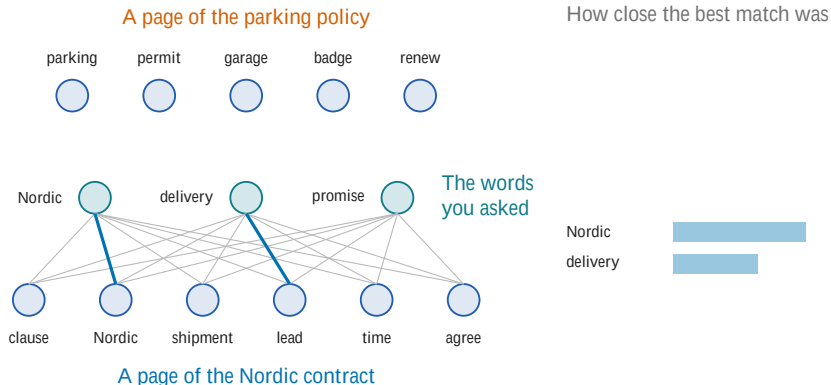


A page of the Nordic contract

Best match, one question word at a time

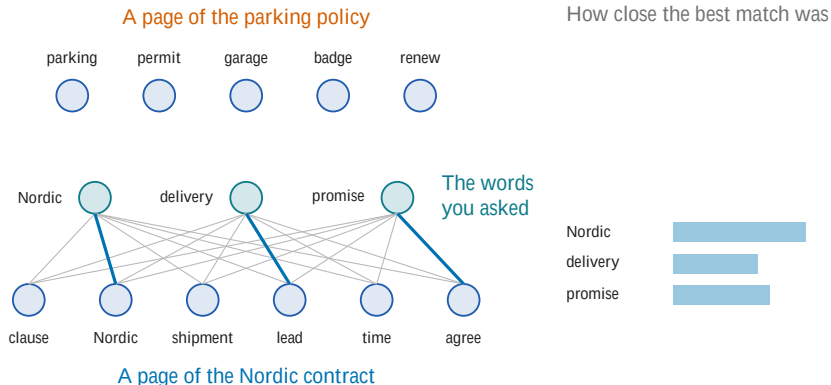


Best match, one question word at a time



Illustrative, not measured.

Best match, one question word at a time



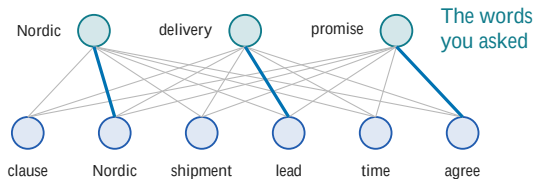
Illustrative, not measured.

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parking permit garage badge renew

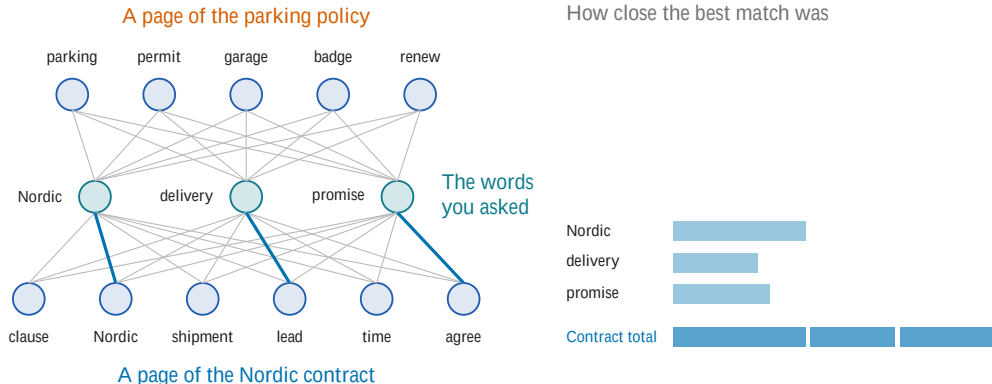
How close the best match was



A page of the Nordic contract

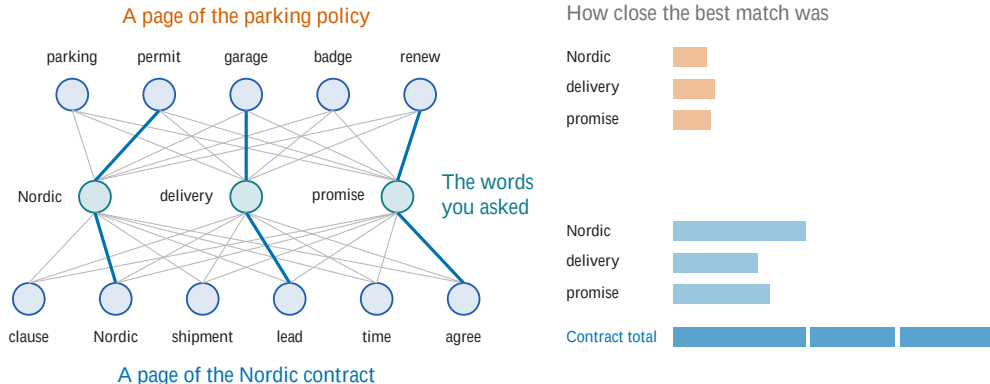
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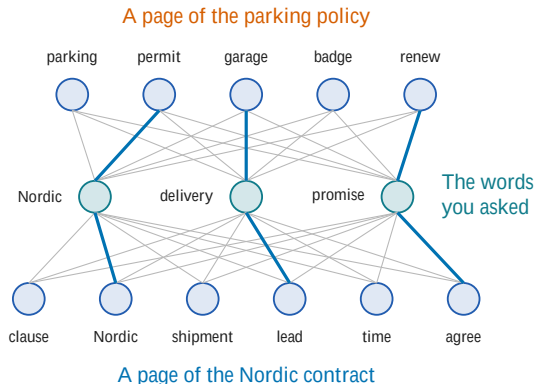
Illustrative, not measured.

Best match, one question word at a time



Illustrative, not measured.

Best match, one question word at a time



How close the best match was

Nordic



delivery



promise



Parking total



Nordic



delivery



promise



Contract total



Illustrative, not measured.

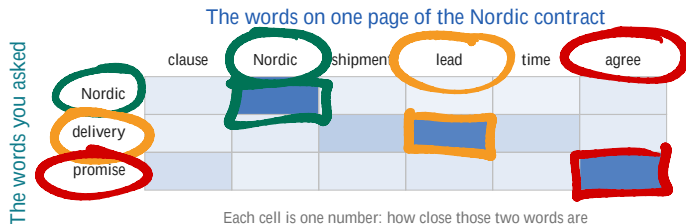
What the best match score actually computes

The words you asked

The words on one page of the Nordic contract

	clause	Nordic	shipment	lead	time	agree
Nordic						
delivery						
promise						

What the best match score actually computes



What the best match score actually computes

The words you asked

The words on one page of the Nordic contract

	clause	Nordic	shipment	lead	time	agree
Nordic						
delivery						
promise						

Each cell is one number: how close those two words are

What the best match score actually computes

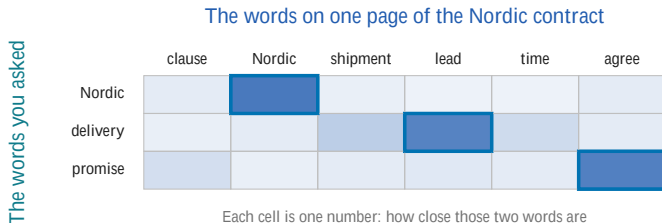
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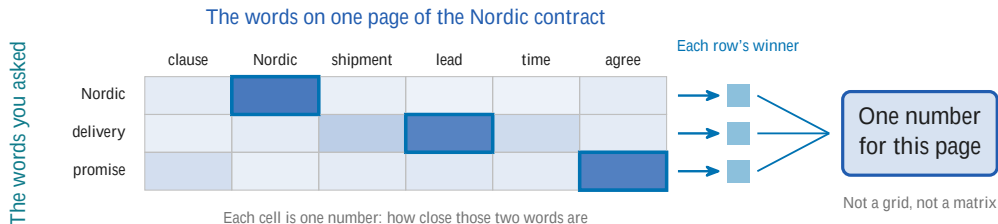
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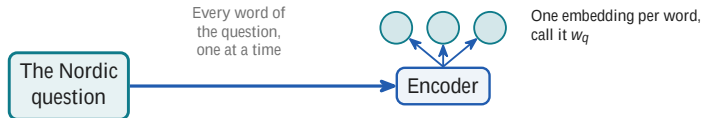
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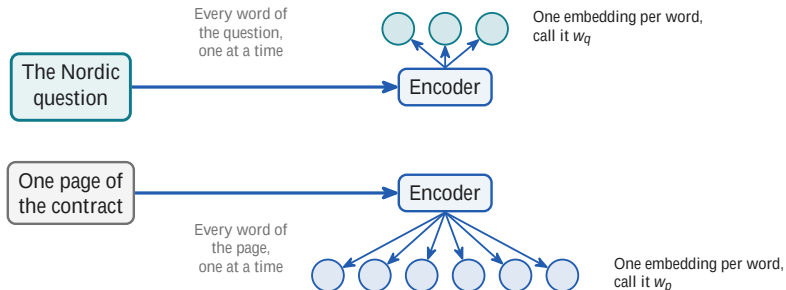


The best match score



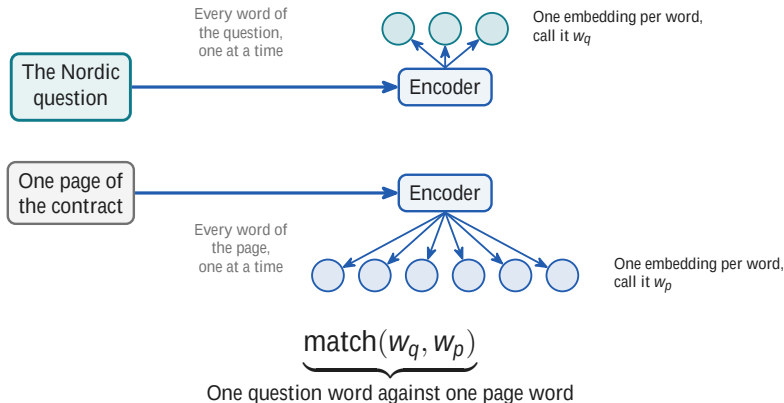
CoBERT, SIGIR 2020.

The best match score



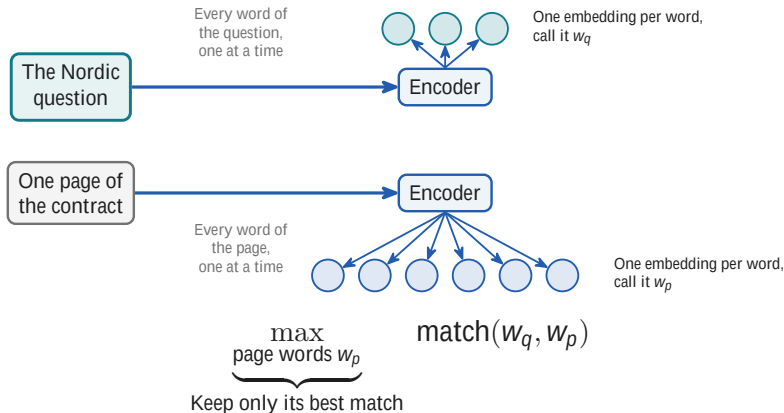
CoBERT, SIGIR 2020.

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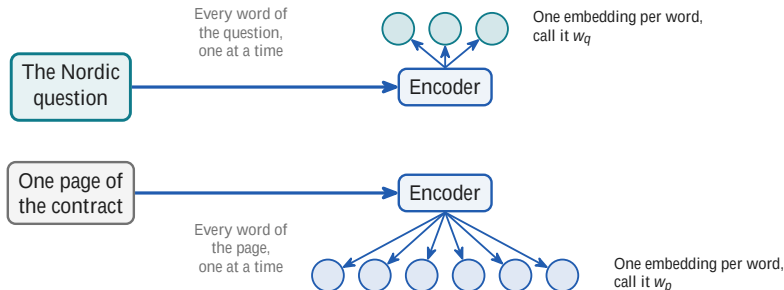
CoBERT, SIGIR 2020.

The best match score



CoBERT, SIGIR 2020.

The best match score



$$\text{score} = \sum_{\substack{\text{question words } w_q \\ \text{Every word you asked}}} \max_{\substack{\text{page words } w_p \\ \text{Its best match}}} \text{match}(w_q, w_p)$$

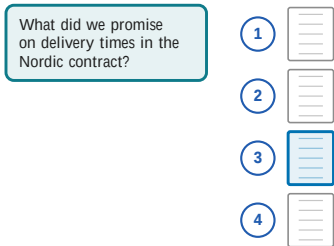
ColBERT, SIGIR 2020.

Rung 3 of three

3

Hand the whole shortlist to a language model

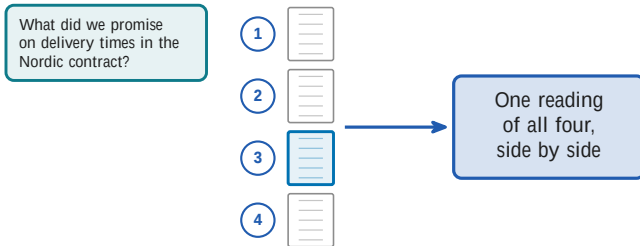
Ask for the order, not for a score



The shortlist, numbered

The model reads the whole shortlist in one prompt and answers with an order, so there is no score left to threshold (RankGPT, 2023).

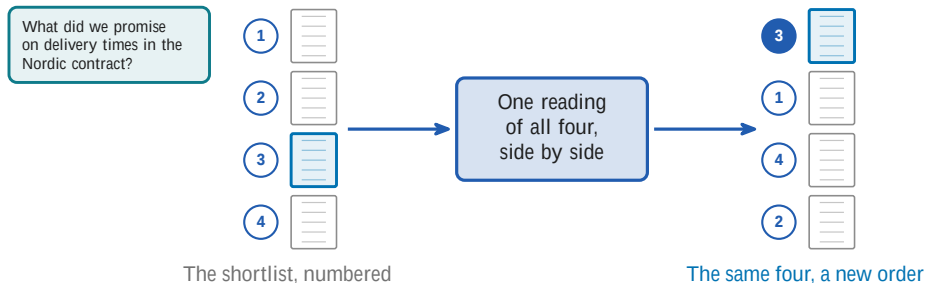
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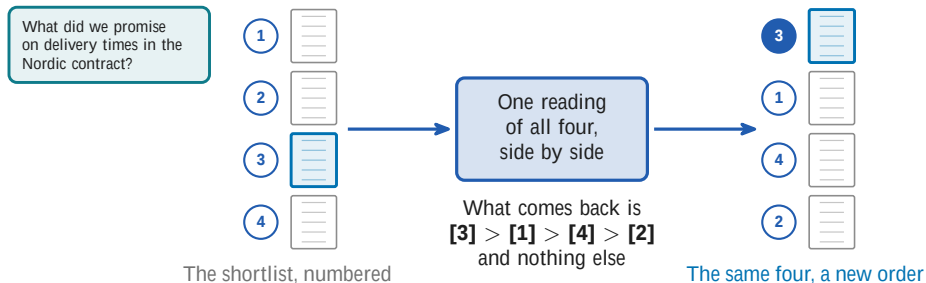
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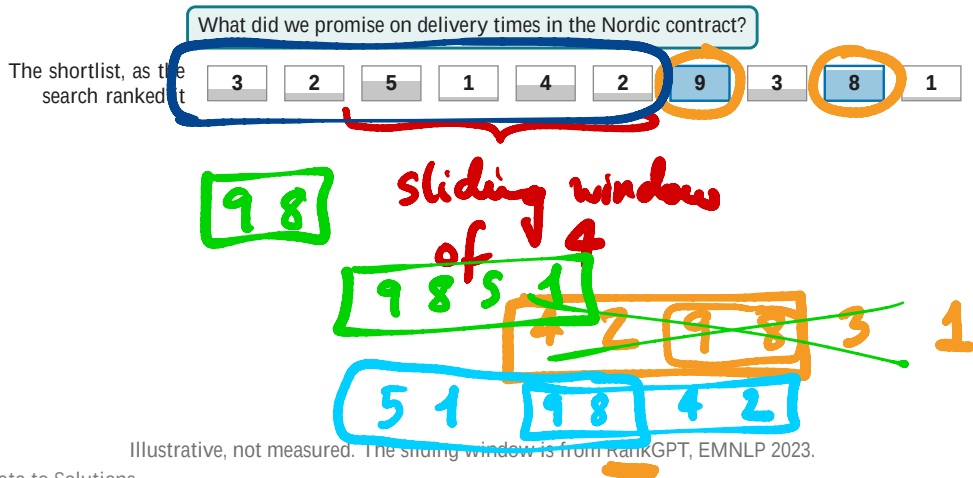
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The window slides from the back to the front

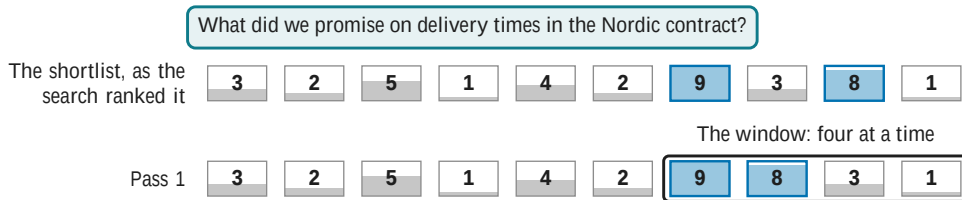
What did we promise on delivery times in the Nordic contract?

Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

The window slides from the back to the front



The window slides from the back to the front



Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

The window slides from the back to the front

What did we promise on delivery times in the Nordic contract?

The shortlist, as the search ranked it



The window: four at a time

Pass 1



Pass 2



Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

The window slides from the back to the front

What did we promise on delivery times in the Nordic contract?

The shortlist, as the search ranked it



The window: four at a time

Pass 1



Pass 2



Pass 3



Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

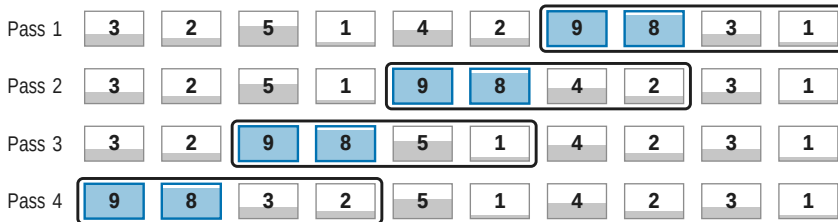
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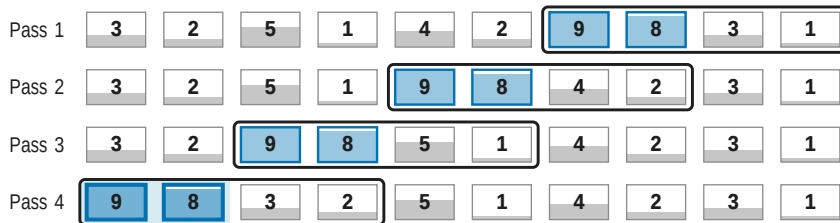
The window slides from the back to the front

What did we promise on delivery times in the Nordic contract?

The shortlist, as the search ranked it



The window: four at a time



The two best have reached the front

Illustrative, not measured. The sliding window is from RankGPT, EMNLP 2023.

The list is longer than the window



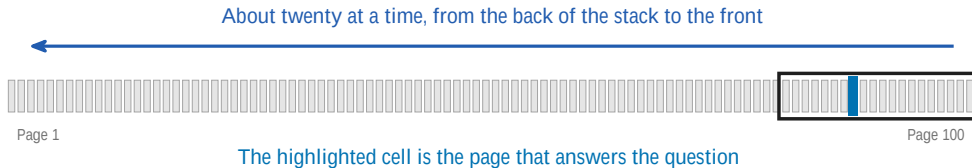
Page 1

Page 100

The highlighted cell is the page that answers the question

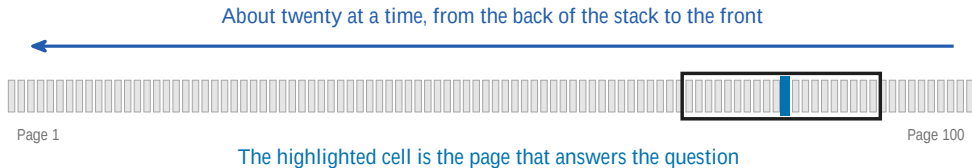
RankGPT: a window of about twenty, stepped ten at a time, so 100 candidates cost ten sequential model calls.

The list is longer than the window



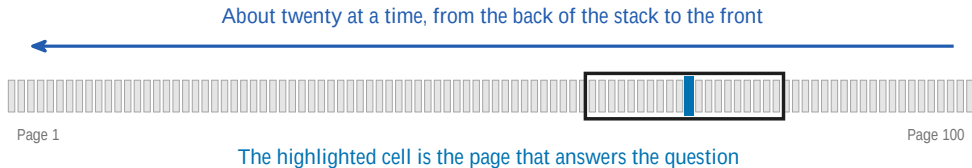
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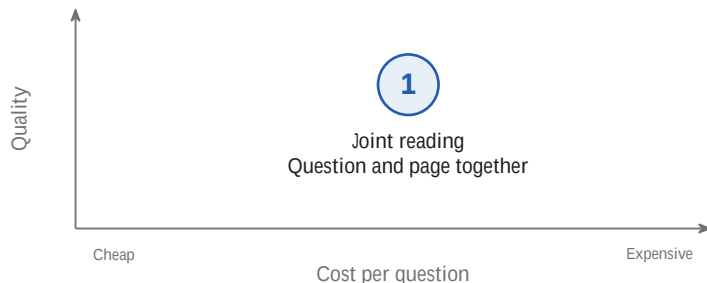
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The three ways on one picture



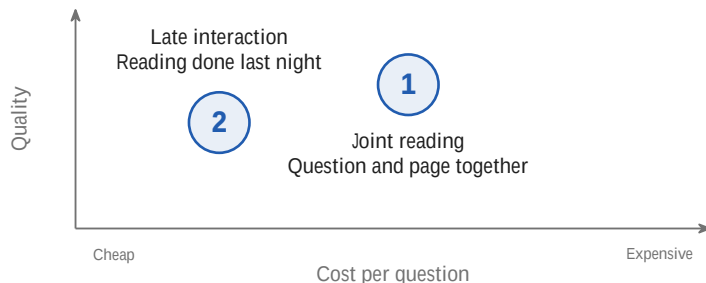
Illustrative, not measured: three papers, three hardware setups, three metrics, no numbers on either axis.

The three ways on one picture



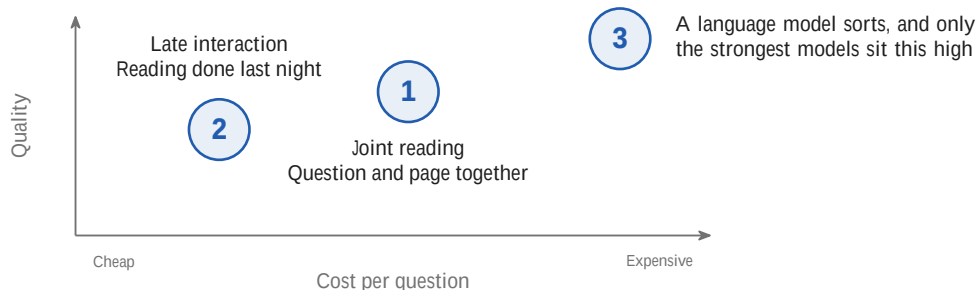
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The three ways on one picture



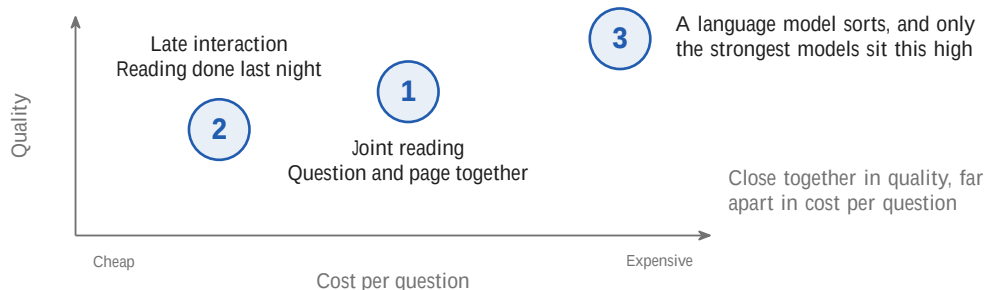
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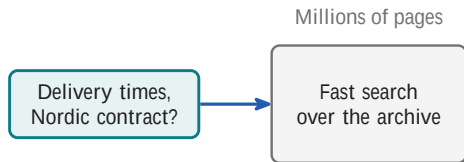


Section 3

What to remember

3

The shape of a real system



The shape of a real system



The shape of a real system



The shape of a real system

